

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
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Section / Batch:	2nd YEAR / CSE	Date:	24/07/2026

Code of Conduct

I niruthika sri prakash (192511294) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: niruthikasriprakash

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

🔗 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarysearch(int arr[],int n,int key)  
2 {  
3     int low=0,high=n-1;  
4     while(low<=high){  
5         int mid=(low+high)/2;  
6         if(arr[mid]==key)  
7             return mid;  
8         else if(arr[mid]<key)  
9             low=mid+1;  
10        else  
11            high=mid-1;
```

Test Input

```
5 12 18 34 45 56 67 78 89 90  
key:45
```

Output

```
Compilation Error:  
C:/CCPPComp/CCPP1/bin/./lib/gcc/  
x86_64-w64-  
mingw32/10.3.0/./../x86_64
```

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack - Balancing Symbols
5 marks

Q6

Singly Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List
5 marks

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

🔗 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop(),stack;  
2 {  
3     if(top==--1)  
4     {  
5         printf("underflow");  
6         return -1;  
7     }  
8     return stack[top--];  
9 }
```

Test Input

```
10 20 30  
top=2  
pop()
```

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_192511294  
RA_6a72fdee2e36a0.57961323\main.c  
:2:1: error: expected identifier  
or '(' before '[' token  
2:1: error: expected identifier  
or '(' before '[' token
```

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];  
int top;  
void init() {  
    top = 0;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int stack[5];  
2 int top;  
3 void init()  
4 {  
5     top=-1;  
6 }
```

Test Input

initialize the stack

Output

```
Compilation Error:  
C:/CCPPComp/CCPP1/bin/./lib/gcc/  
x86_64-w64-  
mingw32/10.3.0/./../x86_64-  
w64-mingw32/bin/ld.exe:  
C:/CCPPComp/CCPP1/bin/./lib/gcc/
```

 Pending manual review

Q4 Stack

Identify the errors in the following snippet.

```
if(ch == ')') {  
    while(stack[top] != ')') {  
        postfix[j++] = pop();  
    }  
    pop();  
}
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor

Your Fixed Code

```
1 if((ch== ' '))  
2 {  
3     while(top!= -1 && stack[top] != '(')  
4     {  
5         postfix[j++]=pop();  
6     }  
7     pop();  
8 }
```

Test

sta

Output

Cor
C:
RA
:1
or

Q5 Stack - Balancing Symbols

In balancing Bracket, you identify the error.

```
if((ch == ']' && stack[top] == '[') ||  
(ch == '}' && stack[top] == '{')) {  
    pop();  
} else {  
    return 0;  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the e

Your Fixed Code

```
1 if(ch== ']' && stack[top]== '(') ||  
2   (ch==']') && stack[top]== '[') ||  
3   (ch== '}' && stack[top]== '{')) {  
4     pop();  
5 } else{  
6     return 0;  
7 }
```

Q6 Singly Linked List

5 marks

What is the issue?

```
temp = head
while temp.next != NULL:
    print(temp.data)
    temp = temp.next
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp=head
2 while temp.next!=null;
3 print(temp.data)
4     temp=temp.next
```

Test Input

stdin input (optional)

Output

🕒 Pending manual review

Q7 Stack

5 marks

Point out the error.

```
POP(stack):
if top == 0:
    print("Underflow")
else:
    return stack[top--]
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 pop(stack)
2     if top==--1:
3 print("underflow")
4     else:
5 return stack[top--]
```

Test Input

stdin input (optional)

Output

🕒 Pending manual review

Q8 Linked List

Identify the errors in the following code

```
while current != NULL:  
    current.next = prev  
    prev = current  
    current = current.next
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while current!=null  
2 next=current.next  
3 current.next=prev  
4 prev=current  
5 current=next
```

Test Input

stdin input (optional)

Output

🕒 Pending manual review

Q9 Linked List

What is the error in the below code?

```
new.next = head
```

```
head = new
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new=node(data)
2   new.next=head
3   head=new
```

Q10 Queue

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
if front == -1:  
    print("Underflow")  
else:  
    return Q[front++]
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor.

Your Fixed Code

```
1 dequeue(q) :  
2 if front == -1:  
3     print("underflow")  
4     else:  
5         item = q[front]  
6         front = front + 1  
7         if front > rear  
8             front = -1  
9             rear = -1  
10        return item
```

Test

Output

Q1 Stack

Point out the error?

PEEK(stack):

return stack[top+1]

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor.

Your Fixed Code

```
1 peek(stack) :  
2 return stack[top]
```

Q2 Circular Queue

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor.

Your Fixed Code

```
1 if((rear+1) % max == front) {
2     printf("queue full");
3 }
4 else:
5 rear=(rear+1) % max ;
6 queue[rear]=vehicle;
```

Tr

C

Q3 Stack

Point out the error.

if top == MAX:

Overflow

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the box below.

Your Fixed Code

```
1 if top==max-1:  
2 print("overflow")
```

Q4 Stack

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor.

Your Fixed Code

```
1 pop() :  
2 if top == -1
```

Q5 Doubly Linked List

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {  
    if (del->prev != NULL)  
        del->prev->next = del->next;  
    if (del->next != NULL)  
        del->next->prev = del->prev;  
}
```

 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the box below.

Your Fixed Code

```
1 void deletenode(node*del) {  
2     if (del->prev!=null)  
3         del->prev->next=del->next;  
4     else  
5         head=del->next;  
6     if(del->next!=null)  
7         del->next->prev=del->prev;  
8 }
```

Q6 Queue

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor.

Your Fixed Code

```
1 void enqueue(int data)  
2 {  
3     if(rear==max-1) return;  
4     queue[rear++]=data;  
5 }
```

Q7 Queue - Deque

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor.

Your Fixed Code

```
1 void insertfront(int data) {  
2     if (front == -1) {  
3         front = rear = 0;  
4     }  
5     else if (front == 0) {  
6         front = max - 1;  
7     }  
8     else {  
9         front = front - 1;  
10    }  
11    deque[front] = data;  
12 }
```

Test

Submit

Output

Binary Search

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

🔗 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the

Your Fixed Code

```
1 int binarysearch(int arr[],int l,int r,int x)  
2 {  
3     if(l>r) return -1;  
4     int mid=l+(r-1)/2;  
5     if(arr[mid]==x) return mid;  
6     if (arr[mid]>x)  
7         return binarysearch(arr,l,mid-1,x);  
8     return binarysearch(arr,mid+1,r,x);  
9 }
```

Q9 Circular Queue

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor.

Your Fixed Code

```
1 int dequeue() {  
2     if(front==-1) return -1;  
3     int data=queue[front];  
4     if(front=rear)  
5         front=rear=-1;  
6     else  
7         front=(front+1)% max;  
8     return data;  
9 }
```

Q10 Stack - Postfix Expression

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor.

Your Fixed Code

```
1 int op1=pop();  
2 int op2=pop();  
3 result=op2 - op1; //for subtraction  
4 push(result);
```